

CSCD602: Advanced Software Engineering

Capstone Project Coursework

Assessment: Group Capstone Project – Individual Submission

Submission Platform: SAKAI LMS

Project Type: Complete Software Development Project

Assignment

Working in self-selected groups, identify and agree on a software project topic that addresses a genuine problem, need, or opportunity. Your group is expected to take the project through the complete software engineering lifecycle, from requirements gathering through implementation, testing, deployment, documentation, maintenance, and future evolution.

The final submission should demonstrate that your group can apply advanced software engineering principles to develop a complete, functional and deployable software application.

1. Project Requirements

Your group must undertake the following stages:

A. Requirements Engineering

- Identify stakeholders and users.
- Gather and analyse requirements.
- Define functional and non-functional requirements.
- Identify system constraints and assumptions.
- Produce a complete Software Requirements Specification (SRS).

B. System Analysis and Design

Provide appropriate design artefacts, including, where applicable:

- System architecture
- Use-case diagrams
- Class diagrams
- Sequence diagrams
- Activity/process diagrams
- Entity-Relationship/Data model
- Component diagrams
- User-interface designs/wireframes
- Database design
- Technology architecture

C. Implementation

Develop the complete working software application, not merely a prototype or mock-up.

The implementation should demonstrate:

- Appropriate programming languages/frameworks
- Proper software architecture
- Database integration where required
- API/service integration where appropriate
- Authentication and authorisation where required
- Appropriate security practices
- Error handling and validation
- Responsive and usable interfaces
- Maintainable and well-structured code

D. Testing and Quality Assurance

Develop and execute an appropriate testing strategy, including relevant tests such as:

- Unit testing
- Integration testing
- System testing
- User acceptance testing
- Functional testing
- Usability testing
- Performance testing, where appropriate
- Security testing, where appropriate

Provide evidence of testing and document identified defects and their resolution.

E. Deployment

The completed application must be made accessible for assessment.

Where appropriate, use platforms such as Vercel or another suitable hosting/deployment platform. Vercel supports production deployments and provides deployment URLs that can be used to access deployed applications.

Each group must provide:

- Live application URL
- Admin URL, where applicable
- Test-user credentials
- Administrator credentials, where applicable
- Source-code repository link
- Any other information required by the examiner to access and test the application

Important: Do not submit passwords or sensitive credentials in publicly accessible source-code repositories.

2. Software Documentation

The group must produce a comprehensive project documentation covering the complete development process.

The documentation should include, at minimum:

1. Project title and group information
2. Introduction/background
3. Problem statement
4. Aim and objectives
5. Stakeholder analysis
6. Requirements gathering and analysis
7. Software Requirements Specification (SRS)
8. System analysis
9. System design
10. Architecture
11. Database/data design
12. User-interface design
13. Implementation
14. Testing and quality assurance
15. Deployment
16. User manual
17. System administration/maintenance guide
18. Security considerations
19. Challenges and solutions
20. Limitations
21. Maintenance strategy
22. Future system evolution/enhancements
23. Conclusion
24. References
25. Appendices

3. Maintenance and Future Evolution

Your project must demonstrate how the software can continue to evolve after deployment.

Discuss:

- Corrective maintenance
- Adaptive maintenance
- Perfective maintenance
- Preventive maintenance
- Bug-fixing processes
- Version control

- Dependency/library updates
- Security updates
- Scalability
- Future features
- Technology migration
- Future integration with other systems
- Possible AI or other emerging-technology enhancements, where relevant

Provide a future evolution roadmap showing how the application could develop beyond the current version.

4. Group Information

Every submission must clearly identify:

Group Number/Name: _____

Project Title: _____

No. Group Member Student ID Major Contribution

- 1
- 2
- 3
- 4
- 5

5. Individual Submission Requirement

Although the software is developed as a group, EVERY GROUP MEMBER MUST SUBMIT THE FINAL PROJECT INDIVIDUALLY THROUGH SAKAI LMS.

Each student should submit the same core group project deliverables but must ensure that their submission clearly identifies their individual contribution and understanding of the system.

The lecturer may require individual students to demonstrate or explain components of the application during assessment.

6. Final Submission Package

Each student must submit through the SAKAI LMS Assignment tool. Sakai's Assignment tool supports individual student submissions and allows instructors to access submitted files for grading.

Required Submission:

1. Complete Project Documentation – PDF
2. SRS – PDF
3. Design Documentation – PDF
4. Testing Report – PDF
5. User Manual – PDF
6. Maintenance & Future Evolution Plan – PDF
7. Source-Code Repository Link
8. Live Application URL
9. Admin/Test Access Details
10. Any additional supporting materials

To make grading easier, groups may combine the documentation into one comprehensive Project Documentation PDF, provided all required sections are clearly identified.

7. Recommended Submission Structure

Each student should submit one compressed folder (.zip) through SAKAI containing:

```
GroupNumber_ProjectName_StudentID/  
|  
├── Project_Documentation.pdf  
├── SRS.pdf  
├── Testing_Report.pdf  
├── User_Manual.pdf  
├── Maintenance_and_Evolution.pdf  
├── README.pdf  
└── Links.txt
```

The Links.txt file should contain:

Group Number/Name:

Project Title:

Live Application:

Admin URL:

Source Code Repository:

Test User:

Username:

Password:

Administrator:

Username:

Password:

8. README Requirement

The README should provide a quick guide for the examiner containing:

- Project overview
- Main system features
- Technology stack
- How to access the application
- Test credentials
- Admin access
- Source-code repository
- Deployment information
- Known limitations
- Any special instructions required for testing

9. Assessment Criteria

Component	Marks
Requirements Engineering & SRS	8
System Analysis & Design	7
Software Implementation	12
Testing & Quality Assurance	5
Deployment & Accessibility	4
Documentation & User Manual	4
Maintenance & Future Evolution	5
Individual Contribution & Understanding	5
Total	50

10. Important Rules

- The project must be fully functional and deployable.
- A static mock-up or non-functional prototype is not sufficient.
- All major claims and design decisions should be properly documented.
- Source code should be organised and maintainable.
- Every group member must understand the overall system.
- Every group member must submit individually through SAKAI LMS.
- The Group Number/Name, Project Title, Member Names and Student IDs must appear clearly in the documentation.
- The live application must remain accessible for grading.
- Groups are responsible for ensuring that all submitted links work.
- Any third-party APIs, services, libraries, datasets or frameworks used must be appropriately acknowledged.